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From Kernels to Standards: How Kokkos and Modern C++ Future-Proof Critical Simulation Codes

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FRONTIER

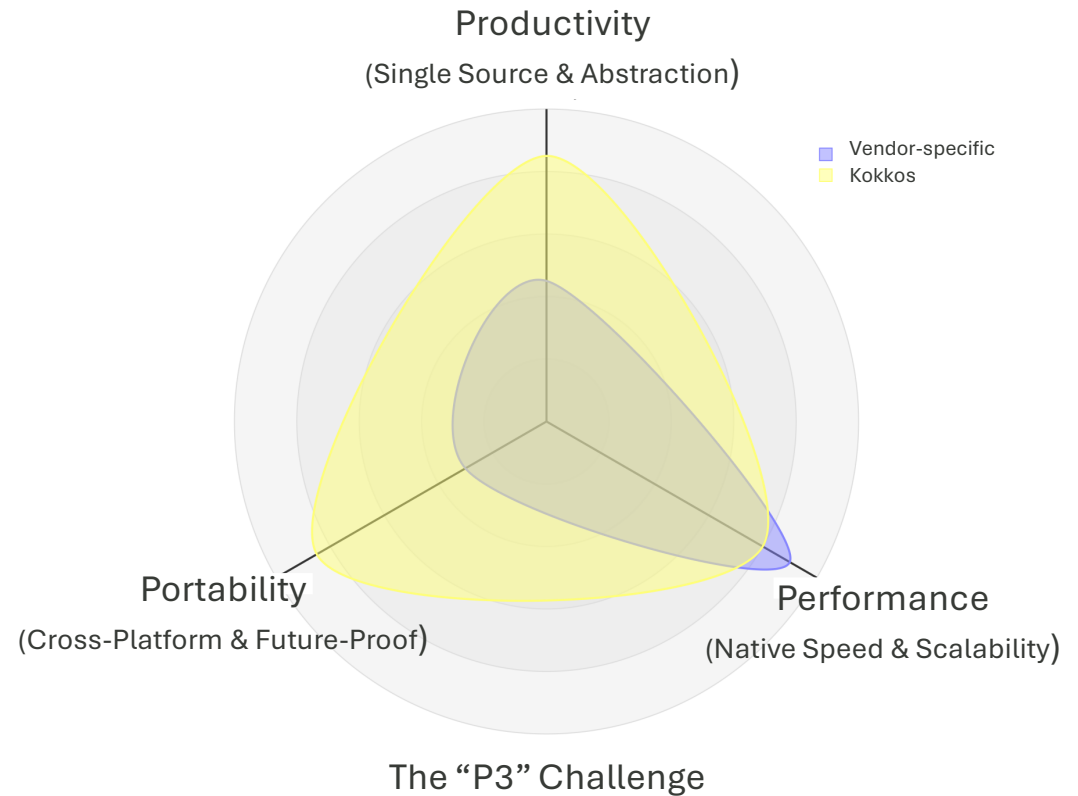
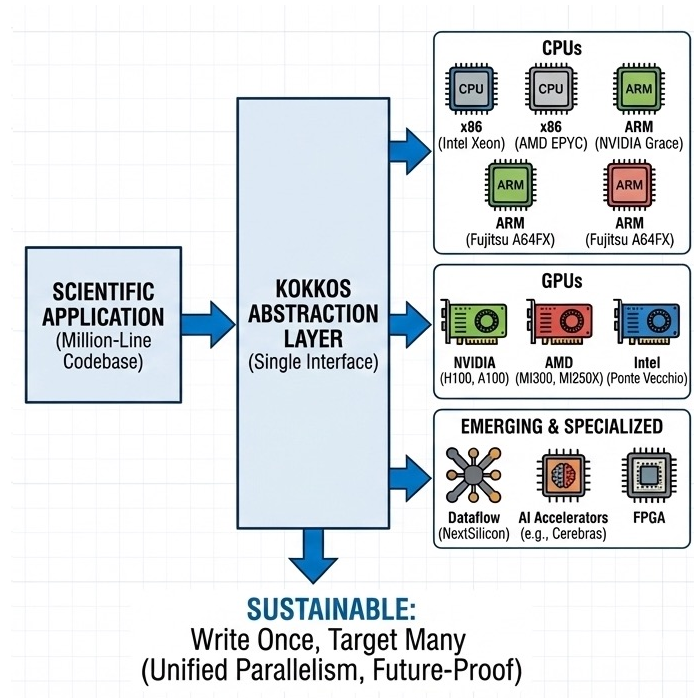


Acknowledgments

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The Exascale Era Challenge



Three Pillars to Tackle the Exascale Challenge



Kokkos



Standard C++



HPSF

HPSF



The "Kernels" – Kokkos and Performance Portability



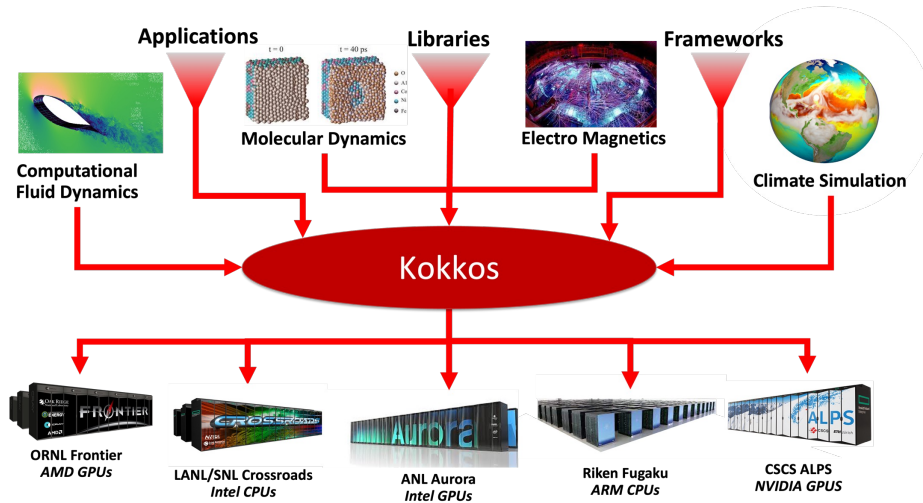
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kokkos

The Kokkos Programming Model



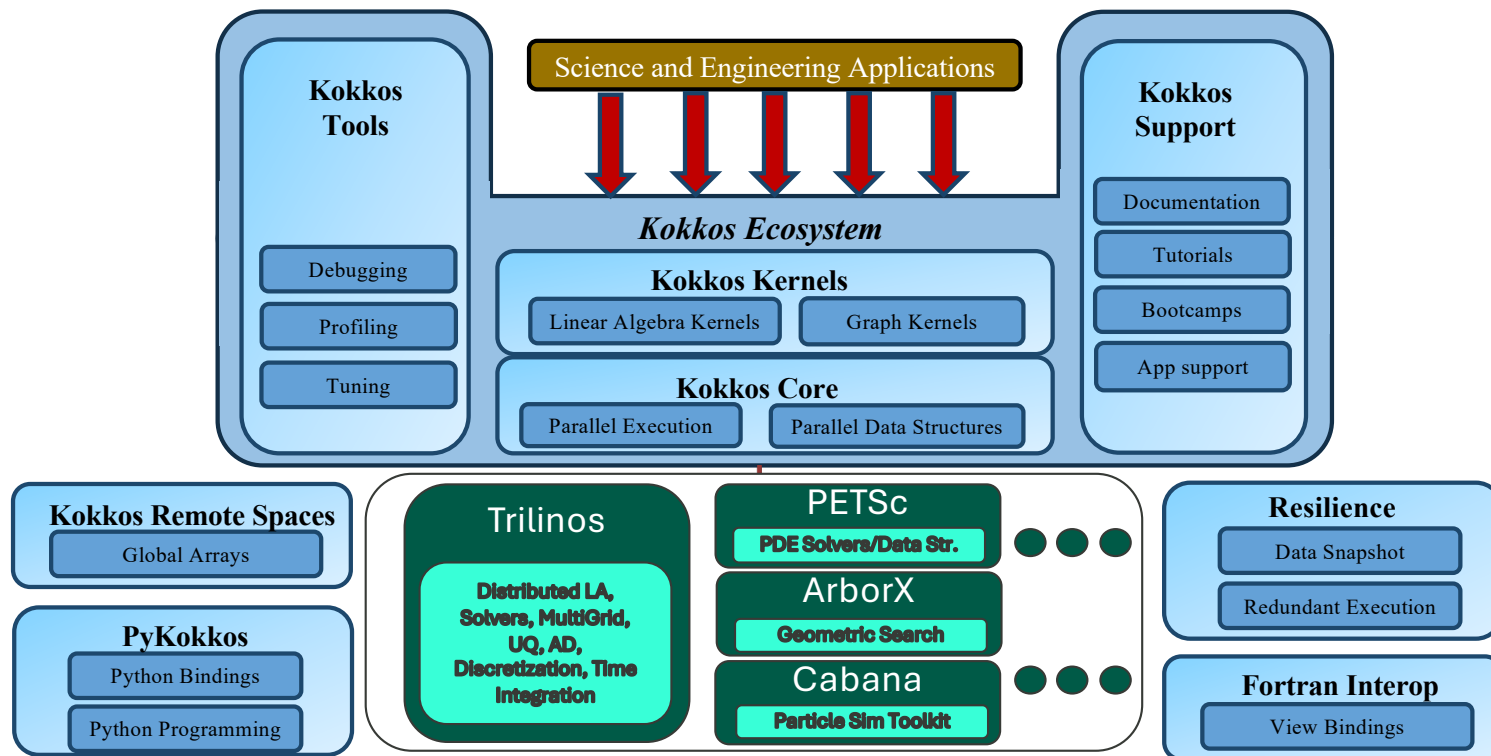
What is Kokkos?

- A C++ Programming Model for Performance Portability
 - Implemented as a template library on top of CUDA, OpenMP, .
 - Aims to be descriptive not prescriptive
 - Aligns with developments in the C++ standard
 - Replaces usage of CUDA, OpenMP, HIP, etc.
 - Use each vendors toolchain!
- Expanding solution for common needs of modern science/engineering codes
 - Math libraries based on Kokkos
 - Tools which enable insight into Kokkos

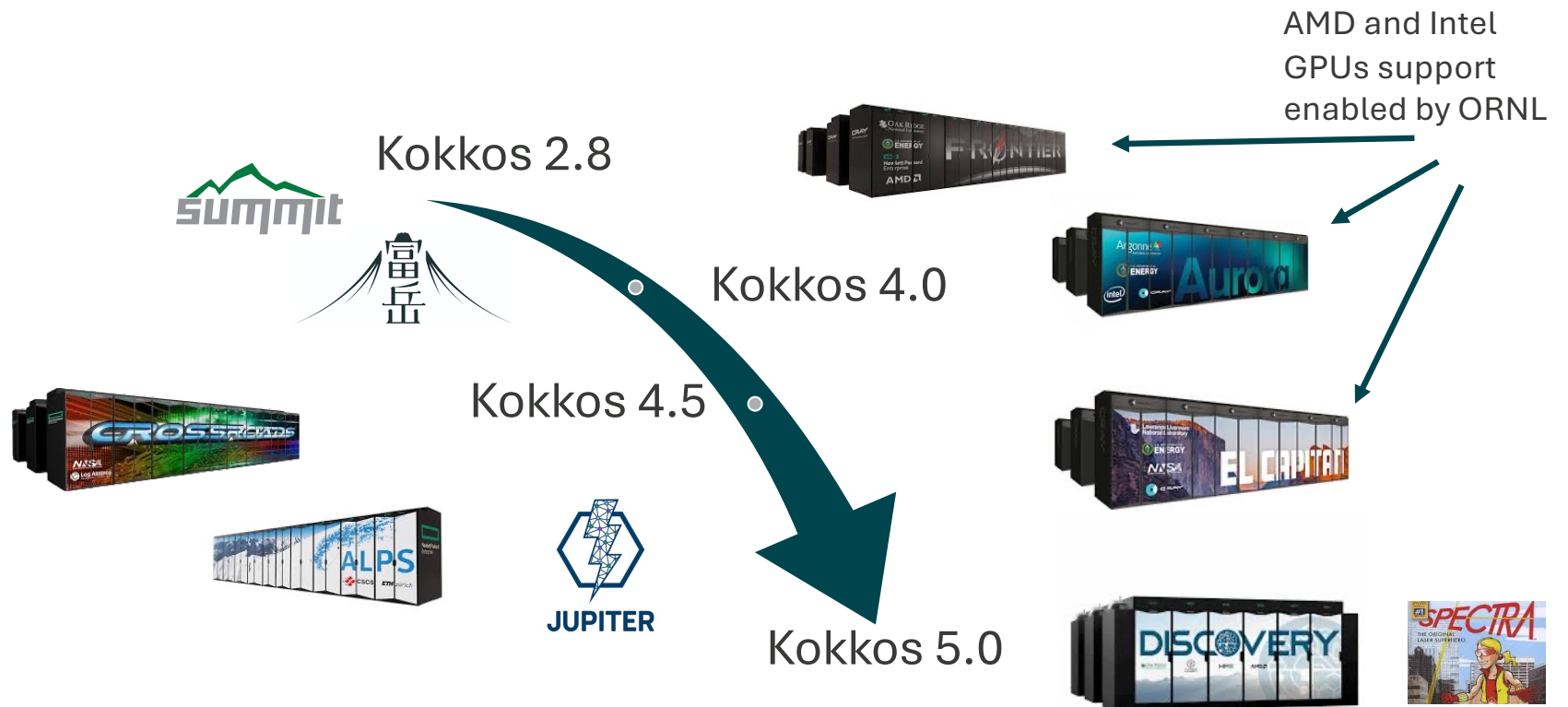
Kokkos is NOT just for GPUs!



More Than Just a Programming Model: A Comprehensive Ecosystem



Powering the DOE Exascale Trilogy



Kokkos at ORNL: Driving Mission-Critical Science

Enabling Libraries	Core Developer Team	Powering Applications
• Geometric Search (ArborX) • Particle-based Simulations (Cabana) • Particle-In-Cell (Picasso) • Mesh Generation (Halyard) • Parallel I/O (ADIOS) • Performance Portability and Fortran Code Porting (YAKL)	5 Dedicated Dual-Role Experts (30%-60% Effort on Kokkos) #1 contributor rank (last 4y) Lead CI and Build & Packaging WGs OLCF support Post-ECP software stewardship funding	• Fusion (VERTEX) • Additive Manufacturing (Toucan, ExaCA, PicassoMPM) • Fracture Mechanics (CabanaPD) • Enrichment (Spinnaker) • Combustion (Quilt, Grit) • Quantum Materials (DMRG++)

We don't just develop Kokkos; we ensure it works for ORNL's mission-critical science

Impact Case Study: ArborX



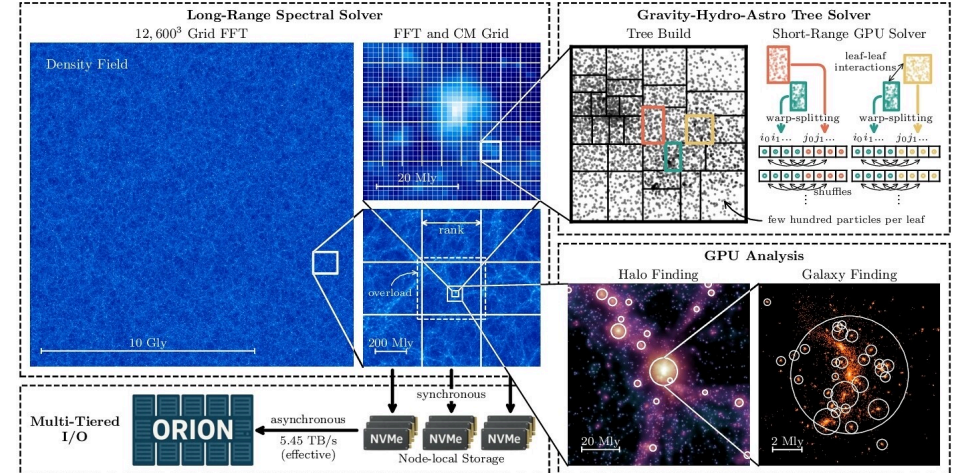
Spatial Indexing and Clustering Library

- Provides data structures to accelerate searching for objects in space
- Includes support for k-NN, ray tracing, collision detection, particle neighbor lists, interpolation, etc.
- Co-designed experimental feature to lower the GPU occupancy

Featured in 2025 Gordon Bell Prize Finalist

“Cosmological Hydrodynamics at Exascale: A Trillion-Particle Leap in Capability”

- Largest ever cosmological hydrodynamics full-sky simulation (> 10x state-of-art)
- Required 9000 nodes for one week on Frontier
- Achieved 513 PFLOPs peak performance
- Produced 100 PB of cosmological data



<https://www.olcf.ornl.gov/2025/11/14/largest-ever-universe-simulation-up-for-supercomputings-highest-prize/>
<https://doi.org/10.1145/3712285.3771786>

Building a Community

Developer Team

Primary
Teams



Sandia
National
Laboratories



OAK RIDGE
National Laboratory



Support
Efforts



CSCS



BERKELEY LAB



LIÈGE
université



TEXAS
The University of Texas at Austin

Helped us get there:



EXASCALE
COMPUTING
PROJECT

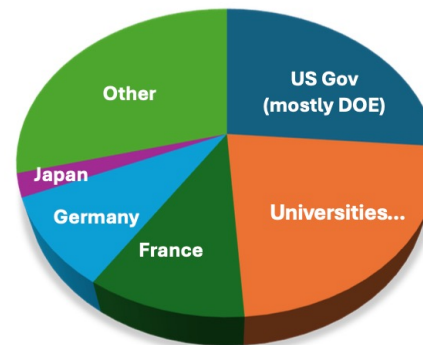
Supports us in our journey to the next level:



Users Community

- Thousands of users (2500 registered on Slack)
- Hundreds of institutions and projects

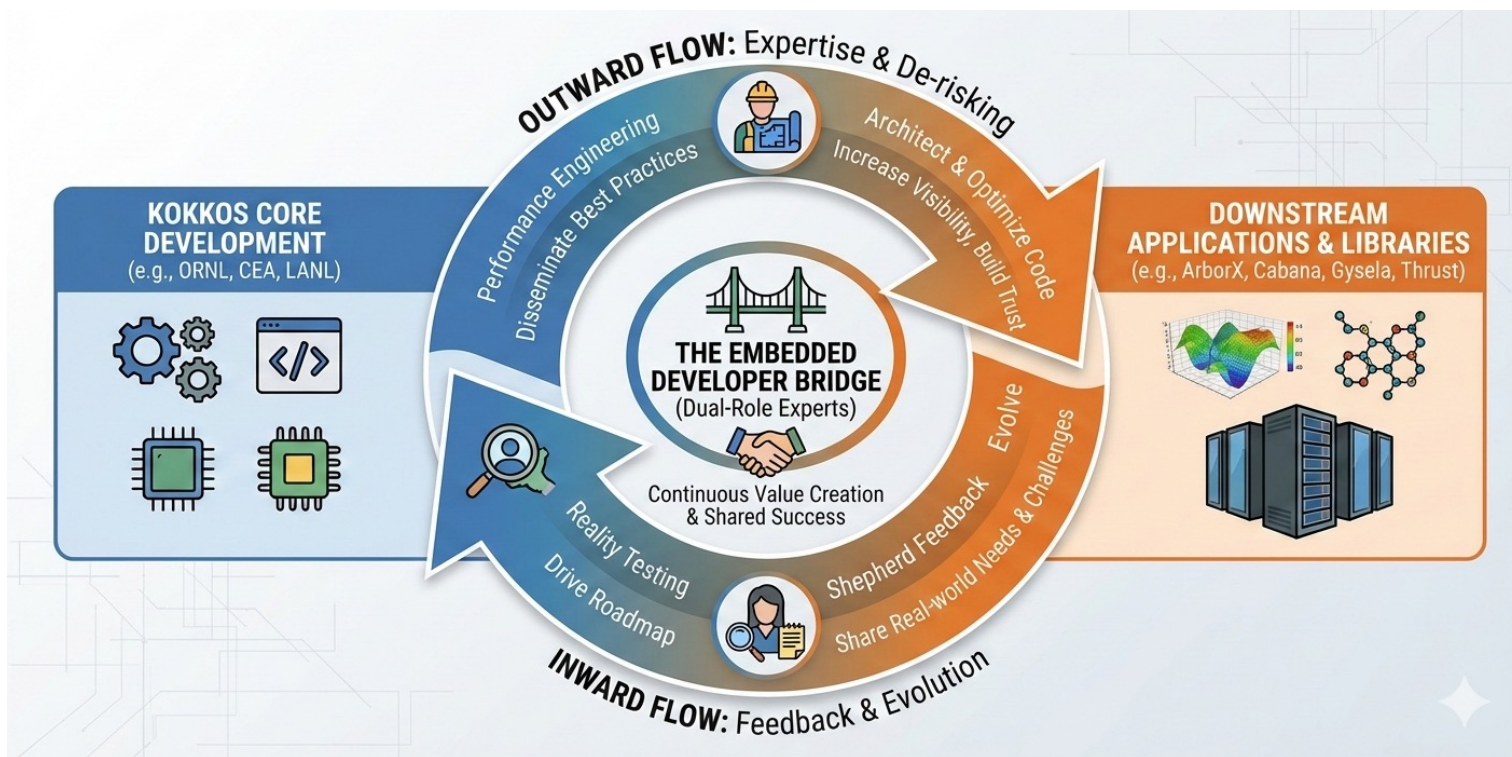
Slack User Origin



HPSF

The Embedded Developer Model

Maximizing Kokkos Return On Investment



Hardening the Ecosystem: Production-Grade Reliability

Structural Governance: Investing in the Future

- **Dedicated Working Groups:** Established “CI” and “Build & Packaging” WGs to move beyond "ad-hoc" maintenance and toward institutionalized quality standards.
- **Reinvestment Rule:** Dedicating 10% of core effort to managing technical debt and paying down the principal
- **Long-term Stability:** Ensuring the core remains agile to be ready for what’s coming next

Rigorous Testing: Integration at Scale

- **Ecosystem Integration:** We don't test in a vacuum; we run continuous integration with downstream partners (e.g. ArborX ,Trilinos, PETSc, LAMMPS) to catch regressions before they impact the science.
- **Universal Coverage:** Continuous testing across the full landscape:
 - On-Prem: Sandia and ORNL internal resources
 - Leadership Facilities: OLCF, ALCF, and NERSC systems

Advanced Tooling: Detecting the Invisible

- **Static Analysis:** Developing Clang-Tidy extensions for invalid code detection and automated migration.
- **Dynamic Safety:** Kokkos Tools recently resolved critical "View of Views" hangs in production code.
- **The Next Frontier:** Increasing investment in tooling to move from bug detection to automated prevention.



The "Standards" – Bridging to ISO C++



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The Strategy of Language Evolution

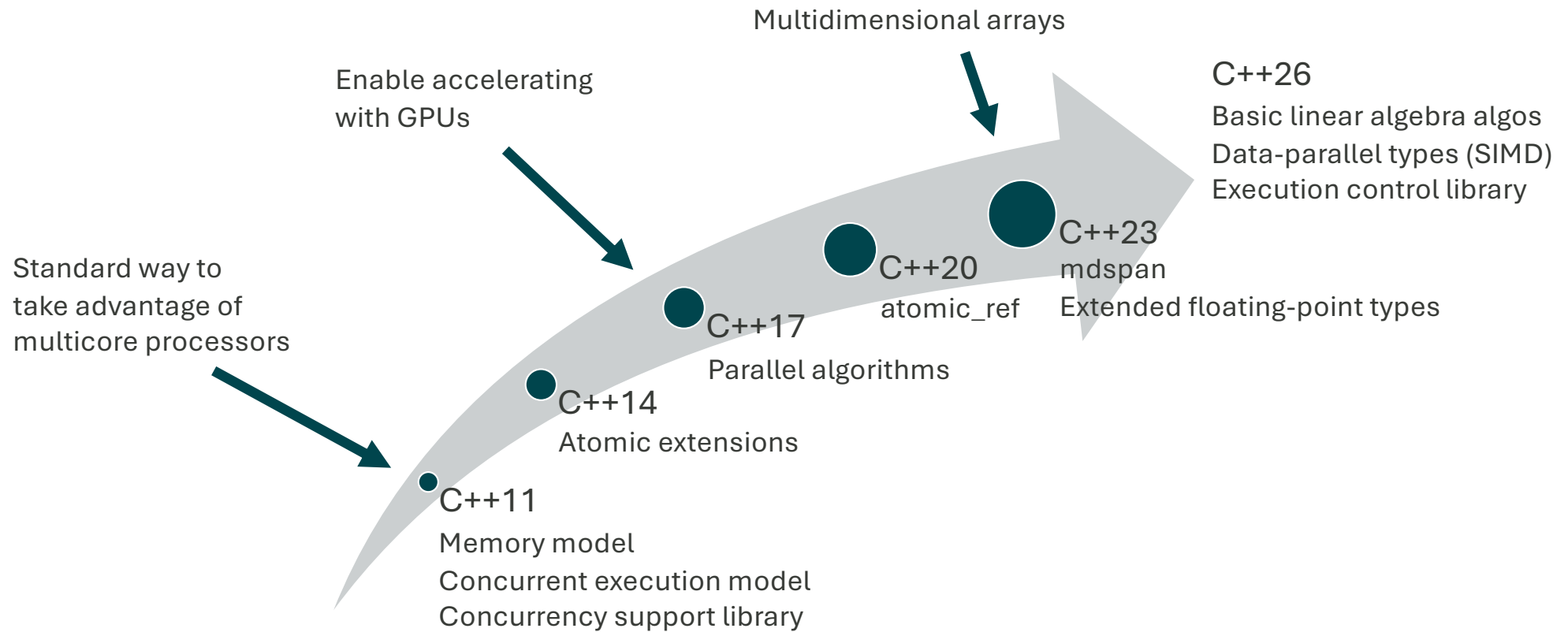
The Problem with Libraries: Even the best libraries represent a dependency. Long-lived codes (30+ years) require foundational stability.

Benefits of Standardization:

- **Vendor Neutrality:** Compilers (GCC, Clang, MSVC, Intel, NVIDIA) must support the standard
- **Interoperability:** Different libraries (e.g., solvers, I/O, viz) can communicate via standard types rather than custom wrappers.
- **Reduced Technical Debt:** Lowering the barrier to entry for new developers by using standard C++ features taught in universities.

Role as DOE lab representative: Ensuring that the unique constraints of HPC are core to C++'s evolution

HPC Features by Standards



Success Story: mdspan (Multi-dimensional arrays in C++23)

```
Kokkos::View<double**> A("A", M, N);  
Kokkos::View<double[4][4], Kokkos::LayoutLeft> B("B");
```

```
std::mdspan A(ptr, M, N);  
std::mdspan<double, std::extents<int, 4, 4>, std::layout_left> B(ptr);
```

```
template <  
  class DataType  
  [, class LayoutType]  
  [, class MemorySpace]  
  [, class MemoryTraits]>  
class Kokkos::View;
```

```
template<  
  class T,  
  class Extents,  
  class LayoutPolicy = std::layout_right,  
  class AccessorPolicy = std::default_accessor<T>  
> class mdspan; (since C++23)
```

Challenge: C++ lacked a native way to represent multidimensional arrays

P0009 mdspan: A Non-Owning Multidimensional Array reference

Impact: Kokkos-inspired feature now allows scientific codes to share data across libraries seamlessly

kokkos/mdspan backport to C++17 deployed in NVHPC
Contributed an implementation to libc++ (LLVM)
Kokkos::View refactored in terms of mdspan in release 5.0

The Next Frontier: Standard Linear Algebra (C++26)

```
dgemv('N', M, N, 1., A, 1, x, 1, 0., y, 1); // 11 parameters      BLAS
KokkosKernels::gemv('N', 1., A, x, 0., y);                      KokkosKernels
std::matrix_vector_product(A, x, y);                            Standard C++
```

```
template<
    [class ExecutionHandle,]
    class InMat,
    class InVec,
    class OutVec>
void KokkosKernels::gemv (
    [const ExecutionHandle& exec,]
    const char trans[],
    typename InMat::const_value_type& alpha,
    InMat A,
    OutMat x,
    typename OutVec::const_value_type& beta,
    OutVec y);
```

```
template<
    [class ExecutionPolicy,]
    InMatrix InMat,
    InVector InVec,
    OutVector OutVec
> void matrix_vector_product( [ExecutionPolicy&& exec,]
                             InMat A,
                             InVec x,
                             OutVec y ); (since C++26)
```

Scientific computing relies on BLAS/LAPACK, but calling these often requires archaic Fortran interfaces or vendor-specific wrappers

P1673: A C++ Standard Library Interface for Parallel Linear Algebra

What it Delivers:

- A modern, type-safe interface for vector and matrix operations
- Support for execution policies (running on GPUs/Multi-core via `std::execution`) and mixed precision
- A common language for developers to express mathematical intent that vendors can optimize under the hood

Strategic Value: Future-proofs computationally expensive parts of simulation codes by making linear algebra a first-class citizen of the C++ language

Leading the HPC Voice in the C++ Community

Serving as "HPC Filter" in WG21: Review incoming proposals to ensure they don't inadvertently harm performance or ignore the needs of large-scale distributed systems

Community Liaison: Bridging the gap between lab scientists and the C++ community at large

- Scientific Computing Track at CppCon
- Disseminate best practices and share lessons learned from other domains (Embedded, Game dev, Trading, etc.)



Goal: Make C++ the most productive language for AI and post-exascale science without losing the "zero-overhead" performance guarantee

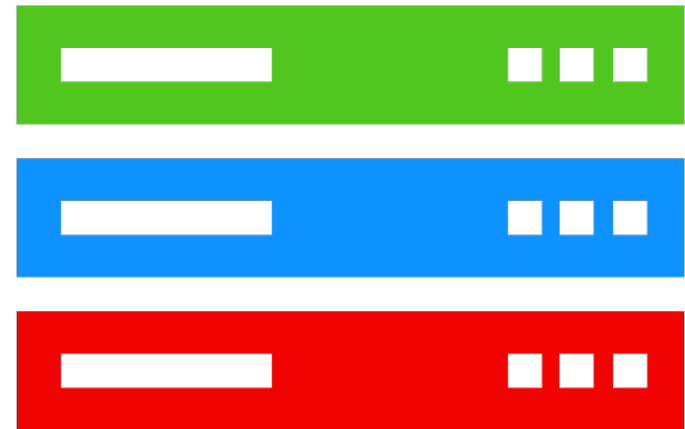


Strategic Future- Proofing – HPSF & Leadership



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HPSF

HIGH PERFORMANCE
SOFTWARE FOUNDATION

Beyond the Code: The High Performance Software Foundation

The Vision:

Creating a neutral, vendor-agnostic home under the Linux Foundation to host and sustain the essential software stack for global HPC.

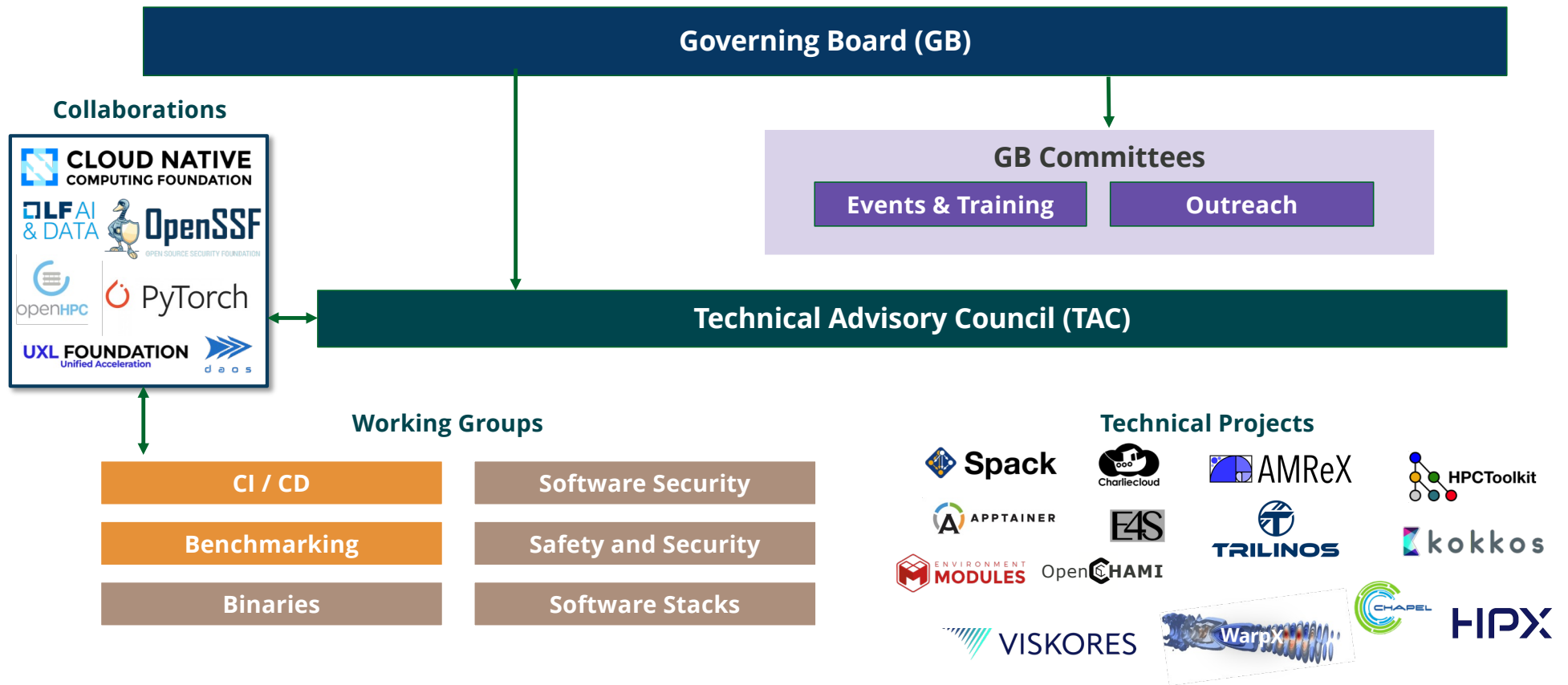
Value to the Labs:

- Ensures that critical tools have a life beyond a single funding cycle.
- Fosters a "neutral ground" for DOE, industry (AMD, NVIDIA, Intel), and international partners to collaborate.

Leadership Role:

- **Co-Founder:** Collaborated with industry and lab partners to define the foundation's core mission and governance.
- **Governing Board Member:** Driving strategic decisions on budget, policy, and cross-foundation resource allocation.
- **Technical Advisory Council (TAC):** Providing technical oversight and ensuring interoperability across flagship projects like Kokkos, Spack, Trilinos, Viskores, HPCToolkit, etc.

HPSF Charter



HPSF Members

Premier



General



Associate



What's in it for Projects?



HPSF

Vendor-Neutral Home & Stability

Ensures project independence and long-term sustainability under the Linux Foundation.

Growth, Branding & Focus

HPSF handles legal, marketing (websites, new logos), and community growth, letting your team focus on the code.

Infrastructure & Automated Testing

Access dedicated hardware and CI/CD Working Group support for robust automated testing and benchmarking.

Collaboration & Events

Host user groups at the annual HPSF Conference and facilitate multi-institutional collaboration and workforce development.

HPSF Impact: Professionalizing the Kokkos Ecosystem

Governance & Operational Maturity

- Formalized project governance and started developing a community playbook.
- Core leadership graduated from the CSCCE Training for Open-Source Leaders (birdaro 2025 program).
- Committed to the OpenSSF Best Practices Program for security and sustainability.

Engineering & Infrastructure

- Secured early access to hardware for pre-release testing, such as NVIDIA Blackwell.
- Implemented LFX Tools and Slack Pro to manage and analyze a community of 2,500+ users.
- Utilizing Sustainable Research Pathways to diversify and grow the developer pipeline.

Transparency & Engagement

- Established regular Release Briefings and recorded technical seminars.
- Institutionalized Kokkos User Group (KUG) meetings and the Tea-Time Seminar Series.

International Strategic Partnerships

Aligning Global Roadmaps: Actively bridging the gap between US, European, and Japanese HPC efforts.

DOE/MEXT (US-Japan): Leading working groups on Programming Languages and Scientific Libraries to ensure our exascale stacks remain compatible and state-of-the-art.

ADAC: Driving international standards for "Portability, Sustainability, and Integration" in large-scale codes.

Strategic Outcome: Reducing redundant development by leveraging global expertise and shared software components.



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CULTURE, SPORTS,
SCIENCE AND TECHNOLOGY-JAPAN



EuroHPC
Joint Undertaking

Cultivating the HPC Community

Expanding the C++ Ecosystem:

- Founded and Chaired the Scientific Computing Track at CppCon
- Created a dedicated space for scientists within the world's largest C++ conference to influence the "mainstream" developer community.

Organizational Service:

- 2nd edition of the HPSF Conference in Chicago March 19-20, 2026
- ISO C++ panel to discuss the latest development of the language and tutorials to teach how to leverage standard parallelism on heterogeneous systems (SC, ISC, HPC Asia)

Mentorship & Workforce Development:

- Engaging with application and other software teams
- Active mentorship of post-docs and junior staff
- Hosting student interns and collaborating with faculties
- Ensuring the "Performance Portability" mindset is passed down to the next generation of computational scientists

Conclusion

Code (Kokkos)

High-performance, portable implementation

Standards (ISO C++)

Language-level stability and longevity

Governance (HPSF/Consortia)

Organizational sustainability and community health

Looking Forward:

- Beyond exascale: AI-integrated HPC and specialized hardware.
- Our commitment: Continuing to bridge the gap between low-level hardware kernels and the high-level needs of the scientific mission.

Thank you – Questions?

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